



Heat transfer aspects of regenerative-cooling in methane-based propulsion systems

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ABSTRACT

In the present article, thermal behavior and heat transfer deterioration (HTD) of transcritical methane as well as the fluid state change in regenerative cooling with straight/curved rectangular channels are studied numerically. Simulations are conducted with a finite-volume based CFD solver utilizing reliable turbulence models and thermo-fluidic relations in transcritical conditions. The experimental and numerical results of hydrogen inside a heated tube in the literature are used for validation. The effects of mass flow rate, outlet pressure, wall temperature, surface roughness, and the channel geometry on the thermal behavior of the coolant fluid are studied in detail. According to the results, variations in the slope of the transport property curves and the inflection point in the density distribution of the coolant flow are proposed as criteria for the recognition of the HTD and transcritical regions, respectively. In addition, the outlet pressure and surface roughness have negligible effects on both methane HTD and transcritical regions in contrast to the channel curvature, mass flow rate, and wall temperature.

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1. Introduction

One of the main challenges of the launch vehicle industry is the high cost of accessing the outer space. Using methane may make future spaceflights less expensive and safer owing to its relatively higher specific impulse, simplicity, reusability, less storage volume, and lower toxicity levels compared to other propellants [1–6]. Thermal load prediction is a critical aspect of methane-based propulsion systems. Regenerative cooling technology is usually exploited as one of the effective cooling methods in liquid propulsion systems [7]. In this technique, one of the propellant fluids passes through cooling channels on the engine wall to absorb heat. In case of cryogenic methane, the coolant may experience Heat Transfer Deterioration (HTD) and severe variations in thermo-physical properties due to the high-temperature gradient in the proximity of the walls, high Reynolds number, and three-dimensional flow structures [8]. Consequently, a comprehensive analysis of the coolant thermal behavior in regenerative cooling channels is of utmost importance to achieve an optimum design, improve the performance, and increase the operating lifetime of propulsion systems [9].

There are few experimental studies [10–13] on the thermal behavior of the cryogenic coolant fluids in transcritical conditions.

These studies focused on the importance of coolant fluid state change and HTD for hydrogen and methane; however, none of them provides further insight on the HTD conditions. In contrast, various CFD methods have been developed to simulate the thermal behavior of the coolant flows at supercritical pressure [6,9,14–23]. Although these methods generally predict the pressure drop and flow properties with acceptable accuracy, they are less precise in estimating the wall temperature that is very important for the thermal analysis of the cooling channels. Moreover, few of these studies [14–20] were devoted to investigating the thermal behavior of transcritical coolant fluids under high heat fluxes and the resulting HTD. In the following, some of the most relevant studies are reviewed.

Wang et al. [14] conducted a comprehensive numerical study on the heat transfer of the methane at supercritical pressure flowing into a horizontal mini-tube and investigated the effect of flow parameters on the Nusselt number variations. They proposed a modified heat transfer expression for the supercritical methane. Further, they showed that the local HTD can occur in the transcritical region under a high heat flux due to the significant wall temperature increment. Urbano and Nasuti [15,16] performed a parametric study on the methane flow at supercritical pressure in an axisymmetric straight heated tube. They discussed the effect of different flow variables such as coolant inlet temperature, pressure, and mass flow rate on the HTD onset. They depicted that the critical wall heat flux value that constitutes an indication of the HTD onset, is linearly proportional to the coolant inlet pressure at

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Nomenclature

A	cross-sectional area	α	Helmholtz free energy
C_p	isobaric specific heat	α^r	residual part of the Helmholtz energy
D	channel diameter	α^0	ideal gas part of the Helmholtz energy
f_w	skin-friction coefficient	δ	reduced density
G_i	constant	ε	roughness
g	gravity acceleration	μ	viscosity
H	total enthalpy	ρ	density
h	convective heat transfer coefficient	$\bar{\tau}_{ij}$	viscous stress tensor
k	thermal conductivity	τ_{ij}	Favre-averaged Reynolds stress tensor
k_r	buoyancy parameter	τ	Inverse reduced temperature
K	turbulence kinetic energy	ρ	density
L_1	unheated length	ω	specific dissipation rate
L_2	heated length		
p	pressure		
Q	mass flow rate		
q	wall heat flux		
q_T	turbulent heat flux		
R	specific gas constant		
T	temperature		
t	time		
u_i	velocity components		
V	velocity magnitude		
x_i	Cartesian coordinates		
X, Y	transverse coordinates		
y^+	non-dimensional wall distance		
Z	stream-wise coordinate		

Subscripts

b	bottom wall
C	critical enhancement term
cr	critical value
e	exit
f	friction term
i	inlet
R	residual term
t	top wall
w	wall
O	dilute gas term
0	stagnation value

a constant mass flow rate. Xu et al. [17] simulated the heat transfer of the turbulent methane flow in different ribbed cooling tubes at a supercritical pressure. They investigated the effects of several geometrical parameters as well as thermal boundary conditions on both heat transfer enhancement and pressure loss. They found that HTD is drastically weakened in a ribbed cooling tube. In another work, Dang et al. [18] numerically investigated the kerosene flow at supercritical pressure in an axisymmetric heated circular tube at various mass flow rates and wall heat fluxes. They showed that HTD may be linked to the near-wall turbulent flow properties and concluded that the critical wall heat flux for the emergence of HTD is linearly related to the mass flow rate at constant pressure. Pizzarelli et al. [19] developed a CFD solver based on a finite-volume method for the thermal analysis of hydrogen and methane at supercritical pressure in a heated tube. Their solver implements the Benedict–Web–Rubin equation of state and transport property relations, and Spalart–Allmaras (S–A) turbulence model. Later, they derived a Nusselt number correlation for HTD of supercritical methane using this solver [20].

In the light of the foregoing discussion, it is revealed that appropriate selection of the thermodynamic and transport equations and turbulence model has a direct impact on the accuracy of coolant fluid flow and thermal simulations. This is because any change in the state of the fluid or the onset of HTD depends on various parameters such as the channel geometry, the amount of wall heat flux, coolant fluid mass flow rate, inlet pressure and turbulent intensity of the flow. Most previous numerical studies have focused on the HTD of the transcritical methane flow inside circular cooling channels. Whereas, the actual engine cooling channels are typically rectangular, curved, and heated asymmetrically [21–23]. As a result, a three-dimensional analysis for the turbulent flow of the transcritical coolant and heat transfer is required. In addition, to the best of authors' knowledge, variations of the transport properties have not been investigated accurately along

the cooling channel at near-critical conditions and deterioration of heat transfer.

In this article, the equations of state and the transport relations are presented for estimation of the thermo-physical properties of transcritical methane and hydrogen. A finite-volume based computational fluid dynamic (CFD) solver is employed to solve the governing equations. The thermal behavior of the methane coolant flow in straight and curved rectangular channels under asymmetric heating and hydrogen in a straight tube was simulated in transcritical condition. The experimental data of Ref. [11] and numerical results of Ref. [20] were used to evaluate the utilized turbulent models and thermo-fluidic relations, respectively. The main contribution of this paper is to simulate and study the HTD in transcritical methane flow as well as the fluid state change inside a straight/curved rectangular cooling channel utilizing reliable turbulence models and thermo-fluidic relations. In addition, variations of the transport properties and density distribution of the coolant flow are proposed as criteria for the recognition of the HTD and transcritical regions, respectively. Moreover, the effects of mass flow rate, outlet pressure, wall temperature, surface roughness, and the channel geometry on the thermal behavior of the coolant fluid (i.e. HTD and fluid state change) are investigated.

2. Thermo-fluidic properties

Methane enters the cooling channel at supercritical pressure and subcritical temperature and exits the channel at supercritical temperature due to the heat absorption. Critical pressure, temperature, and density for methane are 45.9 bar, 190.5 K, and $162.7 \frac{\text{kg}}{\text{m}^3}$, respectively. In such cases, coolant fluid undergoes a pseudo-phase change from a liquid-like state to a gas-like state by passing over the critical temperature. Further, the thermodynamic parameters (e.g. isobaric specific heat and sound speed) experience serious changes in the near-critical region [19]. Hence, the main step in analyzing the coolant thermo-fluidic behavior in transcritical condition is the investigation of the equations of state and transport

Table 1
Proposed thermodynamic and transport property models for transcritical methane and/or hydrogen.

Reference	State equation	C_p	k	μ
Kunz and Wagner [24]	✓			
Span and Wagner [25]	✓			
Setzmann and Wagner [26]	✓	✓		
Friend et al. [27]	✓	✓	✓	
Younglove and Ely [28]	✓	✓	✓	✓
Vogel et al. [29]				✓
Olchoway and Sengers [30]			✓	
Quinones-Cisneros and Deiters [31]				✓
Jaeschke and Schley [32]		✓		

property relations. According to the previous studies [24–32], some of the thermodynamic and transport property models for accurate simulation of transcritical methane and/or hydrogen are listed in Table 1. The relevant references to the equations of state, dynamic viscosity, thermal conductivity, and isobaric specific heat are given as well.

In this study, the operating conditions are less than 90 bar and 600 K for pressure and temperature, respectively. Based on the wide range of the thermodynamic conditions of methane and the required accuracy, the equation of state presented by Kunz and Wagner [24], the isobaric specific heat presented by Jaeschke and Schley [32], thermal conductivity described by Younglove and Ely [28], and viscosity by Quinones-Cisneros and Deiters [31] are adopted. The equation of state by Kunz and Wagner [24] is defined as follows:

$$\alpha(\delta, \tau) = \alpha^0(\delta, \tau) + \alpha^r(\delta, \tau) \quad (1)$$

where $\alpha = \frac{a}{RT}$ stands for the dimensionless form of the specific Helmholtz energy, which is a function of temperature (T) and density (ρ). α^0 and α^r represent the properties of the ideal-gas and the behavior of real gas at a given density and temperature, respectively that can be found in [22]. α^r denotes the residual part of the Helmholtz energy which is determined by fitting curves to the experimental data. $\delta = \frac{\rho}{\rho_{cr}}$ and $\tau = \frac{T}{T_{cr}}$ are the reduced density and inverse reduced temperature of the fluid, in which ρ_{cr} and T_{cr} are the critical density and temperature, respectively. Since the Helmholtz energy is one of the four fundamental forms of the state equation, all thermodynamic properties can be calculated by the combination of α^0 , α^r and their derivatives with respect to τ and δ . The discrepancy of state equation for density and enthalpy of Kunz and Wagner are 0.1% and $\pm(0.1 \text{ to } 0.5)\%$ compared to the reference experimental data.

The isobaric specific heat is given as follows [32]:

$$\frac{C_p(\delta, \tau)}{R} = -\tau^2(\alpha_{\tau\tau}^0 + \alpha_{\tau\tau}^r) + \frac{(1 + \delta\alpha_{\delta}^r - \delta\tau\alpha_{\delta\tau}^r)^2}{1 + 2\delta\alpha_{\delta}^r + \delta^2\alpha_{\delta\delta}^r} \quad (2)$$

where $\alpha_{\tau\tau}^0$, $\alpha_{\tau\tau}^r$, α_{δ}^r , $\alpha_{\delta\tau}^r$, α_{δ}^r and $\alpha_{\delta\delta}^r$ are derivatives of α^0 and α^r with respect to τ and δ . The discrepancy of this relation is $\pm(1 \text{ to } 2)\%$ compared to the reference experimental data. The thermal conductivity of Younglove and Ely [28] and the dynamic viscosity of Quinones-Cisneros and Deiters [31] can be expressed as Eqs. (3) and (4).

$$k(\delta, \tau) = k_0(\tau) + \Delta k(\delta, \tau) + k_c(\delta, \tau) \quad (3)$$

$$\mu(\delta, \tau) = \mu_0(\tau) + \mu_f(\delta, \tau) \quad (4)$$

in which k_0 and μ_0 represent the dilute-gas part, Δk is the excess term for the effect of real gas, k_c is the critical enhancement term,

and μ_f is the friction term, taken from [28] and [31]. The difference for viscosity relation of Quinones-Cisneros and Deiters is around $\pm 2\%$.

Base on the selected state equation and transport property relations in the present study, the calculated values for density, isobaric specific heat, viscosity, and thermal conductivity as a function of temperature and for different pressures (6–13 MPa) are presented in Fig. 1. These results are in good agreement with the NIST database [33] for methane as a working fluid.

3. Governing equations

Three-dimensional compressible Favre-Averaged Navier–Stokes (FANS) equations coupled with the energy and state equations are employed for the fluid flow simulations. Flow in actual engine cooling channels is characterized by high Reynolds numbers of the order of 10^7 . Thus, a forced convection heat transfer regime is dominated, which means that buoyancy effects are negligible [16,20]. Herein, the buoyancy parameter introduced by Kurganov et al. [34] is used to estimate the influence of buoyancy on heat transfer as follows:

$$k_r = \frac{2gD}{\tilde{u}^2 f_w} \quad (5)$$

where K_r is the buoyancy parameter, g is the gravity acceleration, $\tilde{u}$ is the mass-averaged flow velocity, and f_w is the skin-friction coefficient. According to the Refs. [20,34], buoyancy effects should be accounted for when the buoyancy parameter is $k_r > 4$, whereas in the conditions of the present study the maximum measured value of the buoyancy parameter is about 0.16 for transcritical hydrogen in the circular channel and 0.61 for transcritical methane in the rectangular channel. Therefore, buoyancy effects are neglected because of the small value of measured k_r in all the test cases considered. Neglecting the body forces (i.e. the buoyancy effects), the fundamental equations of mass, momentum, and energy conservation in Cartesian tensor notation are [35]:

$$\frac{\partial \bar{\rho}}{\partial t} + \frac{\partial \bar{\rho} \tilde{u}_i}{\partial x_i} = 0 \quad (6)$$

$$\frac{\partial \bar{\rho} \tilde{u}_i}{\partial t} + \frac{\partial (\bar{\rho} \tilde{u}_i \tilde{u}_j)}{\partial x_j} = -\frac{\partial \bar{p}}{\partial x_i} + \frac{\partial}{\partial x_j} (\bar{\tau}_{ij} + \tau_{ij}) \quad (7)$$

$$\begin{aligned} \frac{\partial}{\partial t} (\bar{\rho} \tilde{H}) + \frac{\partial}{\partial x_j} \left(\bar{\rho} \tilde{u}_j \tilde{H} + q_{Tj} - k \frac{\partial \bar{T}}{\partial x_j} \right) \\ = \frac{\partial \bar{p}}{\partial t} + \frac{\partial}{\partial x_j} (\tilde{u}_i \bar{\tau}_{ij} + \tau_{ij} \tilde{u}_i) \end{aligned} \quad (8)$$

in which $\tilde{u}_i$, $\bar{\rho}$, $\bar{p}$, and $\tilde{H}$ are the velocity components, density, static pressure, and total enthalpy, respectively. Double prime ($''$), overbar ($\bar{}$), and tilde ($\tilde{}$) indicate fluctuating, the conventional Reynolds-averaged, and Favre-averaged or density-weighted (e.g. $\tilde{f} = \frac{\bar{\rho} f}{\bar{\rho}}$) quantities, respectively. $\bar{\tau}_{ij}$ is the viscous stress tensor, μ the dynamic viscosity, and k the thermal conductivity. Fourier's law is considered for the heat conduction term in the energy equation. Furthermore, $\tau_{ij} = -\overline{\rho u_i'' u_j''}$ and $q_{Tj} = \overline{\rho u_j'' H''}$ are the well-known Favre-averaged Reynolds stress tensor and turbulent heat flux, respectively, that require modeling. The closure of the governing equations is provided by the turbulence models. In this regard, Spalart–Allmaras (S–A) [36], Standard $K-\omega$ [37], and SST $K-\omega$ [38,39] turbulence models are implemented in the present CFD approach. Governing equations are resolved in combination with the selected state equation and transport property relations.

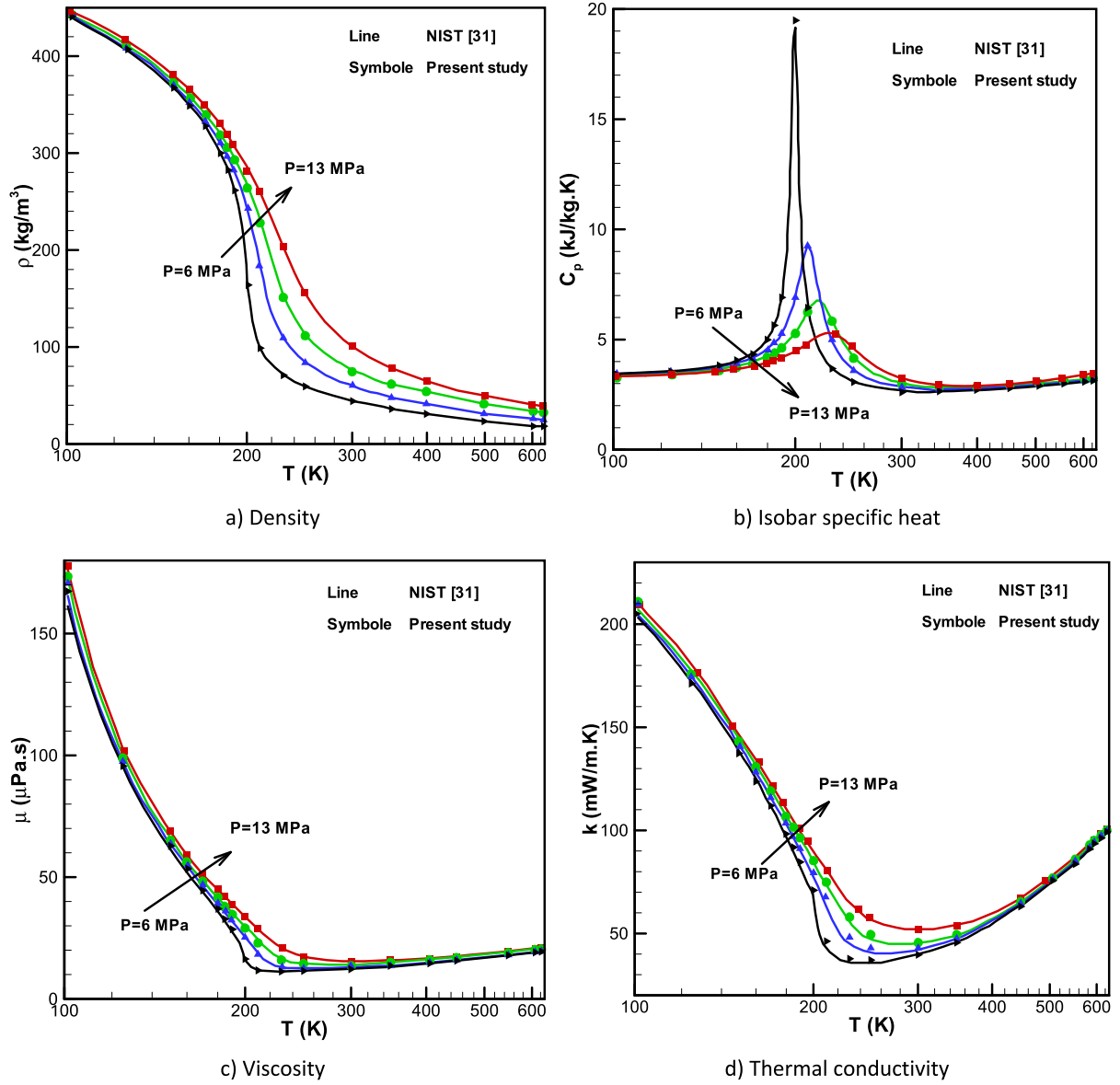


Fig. 1. Calculated thermo-fluidic properties of methane as a function of temperature for different pressures (6, 8, 10, and 13 MPa) and comparison with the NIST database [33].

4. Numerical treatment

The present computations are performed with a finite-volume pressure-based CFD solver. Couple algorithm is used to match pressure and velocity fields. Spatial discretization was preceded using a least squares cell-based algorithm for gradients. The bounded central differencing scheme for momentum and energy is used. Moreover, the second order upwind scheme is adopted for both turbulent kinetic energy and specific dissipation rate in the case of $K-\omega$ models, as well as for turbulent viscosity in the case of $S-A$ model. For the time discretization, the first order implicit scheme is used. The local criterion for numerical convergence (i.e. the maximum relative difference between two consecutive iterations), is less than 10^{-12} for the energy equation. Moreover, direct use of the state equation and the transport property relations for calculation of fluid properties (i.e. ρ , μ , k , and C_p) increases the total computational runtime considerably [40]. Therefore, the values of thermodynamic properties are tabulated and inserted into the CFD solver via linear interpolation. Hence, the fluid properties can be calculated at each computational cell with much less computational effort.

5. Numerical test cases and boundary conditions

5.1. Test Case 1: hydrogen in a heated tube

The experimental data of Ref. [11] on hydrogen as the coolant in a heated tube at transcritical condition is utilized as the validation case for the present numerical simulation. Accordingly, the test case "0681" from [11] is adopted to validate the CFD solution. As shown in Fig. 2, an unheated length (L_1) is considered in order to assure a fully-developed flow field at the entrance of the main tube (L_2). Then, a non-uniform heat flux distribution is imposed to the hydrogen along L_2 . The geometrical parameters and the boundary conditions of the selected test case are sketched in Fig. 3. Here, the mass flow rate is $Q = 0.08$ kg/s, the total temperature of the inlet is set to be $T_0 = 25.056$ K, the static pressure of the outlet is $P_e = 1.613$ MPa and the no-slip condition is applied over the walls. The hydraulic diameter of the tube is 8.5 mm. The hydrogen flow within the tube is assumed to be axisymmetric and steady. The pressure of the coolant fluid inside the channel is at supercritical condition while the temperature varies from a subcritical state to a supercritical state; hence, the system is operating

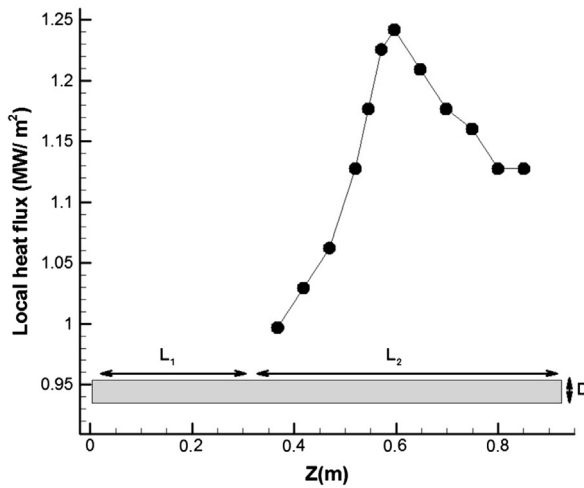


Fig. 2. Geometry and the local heat flux boundary condition of the test case "0681" [11].

in the transcritical condition. According to Fig. 4, the grid convergence study is conducted on different computational grids and an optimum mesh with the size of 32×360 (Radial $\times$ stream-wise) and y^+ of about 1.0 is selected.

5.2. Test Case 2: methane in high aspect ratio rectangular channels

Transcritical methane flow is simulated in three-dimensional high aspect ratio $0.1 \times 0.8 \times 30$ mm straight and curved rectangular channels. The geometry presented in Refs. [21] and [23] is adopted for the straight and curved channels respectively. The boundary conditions of Ref. [21] are used for both cases. Geometry, boundary conditions, and a schematic view of the computational mesh of these channels are shown in Fig. 5. Further, because of the symmetrical geometry and side wall boundary conditions, the coolant flow is simulated in one half of the channel. The total pressure and temperature of the inlet are set to be $P_0 = 9$ MPa and $T_0 = 130$ K, the static pressure of the outlet is $P_e = 7$ MPa, the temperature of the top and bottom walls are $T_b = 600$ K and $T_t = 130$ K, and a linear distribution of temperature is applied to the side wall. Also, the no-slip condition is imposed on all walls. The hydraulic diameter of the channel is 0.17 mm. According to Table 2 and Fig. 6, various computational grids with different resolutions are tested for the mesh independence study and the fine grid with the size of $32 \times 92 \times 100$ (base $\times$ height $\times$ stream-wise) and y^+ less than 1.0 is selected as the optimum mesh.

6. Results and discussion

6.1. Hydrogen in a heated tube

6.1.1. Turbulence modeling and thermo-fluidic relations effects

To investigate the effect of turbulence models on the simulation of transcritical hydrogen flow in Test Case 1, three different

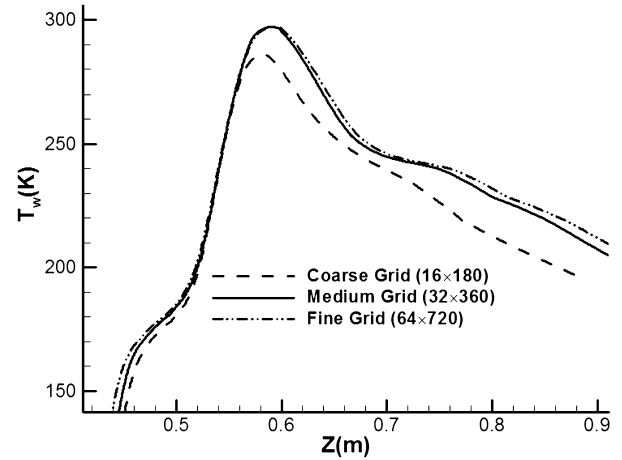


Fig. 4. Wall temperature distribution for different grids.

models (i.e. S-A, $K-\omega$, and SST $K-\omega$) were evaluated. The wall temperature distribution resulted from utilizing these turbulence models are compared to the experimental data of Ref. [11] in Fig. 7. The accuracy of each turbulence model in respect to this experimental data is reported in Table 3 in terms of absolute average deviation (AAD). Fig. 7 shows that the $K-\omega$ model has more deviations from the experimental data in some areas of the cooling channel. This is due to the model sensitivity to the values of ω at the inlet for the boundary layer computations in the internal flows [37]. This deficiency has been removed in the SST $K-\omega$ model by considering cross-diffusion source terms in the ω transport equation [38]. This model uses the Standard $K-\omega$ in near-wall regions and implements the Standard $K-\epsilon$ outside the boundary layer [39]. Therefore, the SST $K-\omega$ model generally has higher accuracy near the wall while being less sensitive to the flow outside the boundary layer. The results of Fig. 7 and Table 3 evidence that the SST $K-\omega$ model offers the highest accuracy and shows the best compatibility with the experimental data. This model is also more accurate in calculating the turbulent viscosity due to the correction of production terms in the turbulent kinetic energy equation and utilization of the transport equation for the turbulent shear stress. On the contrary, the S-A model takes advantage of a transport equation for the turbulent viscosity and is very dependent on the grid size in the near wall region. Consequently, the SST $K-\omega$ model is exploited in the present study to capture the thermo-fluid dynamic of the transcritical coolant flow, although, the S-A model can be used for approximate solutions and the general studies of the coolant flows.

The numerical results of Pizzarelli et al. [20] are used to investigate the effects of implemented state equation and transport property relations on the simulation of transcritical hydrogen flow for the Test Case 1. Pizzarelli et al. employed the S-A turbulence model, the Benedict-Webb-Rubin equation of state and the transport property relations of Younglove and Ely [28] for their simulations. However, in the present study, the state equation of Kunz and Wagner [24], the transport property relations of Jaeschke and

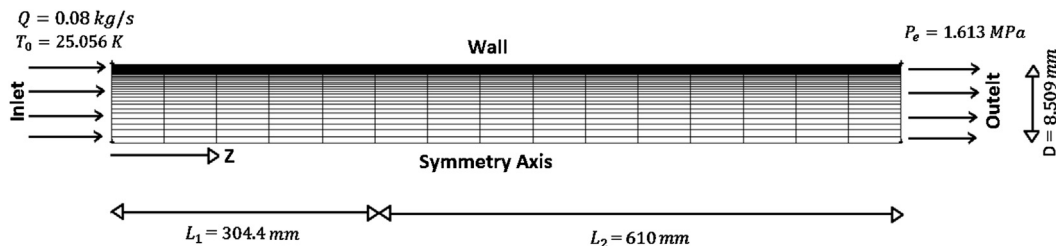


Fig. 3. Computational mesh and boundary conditions of the test case "0681" [11] (figure is not scaled).

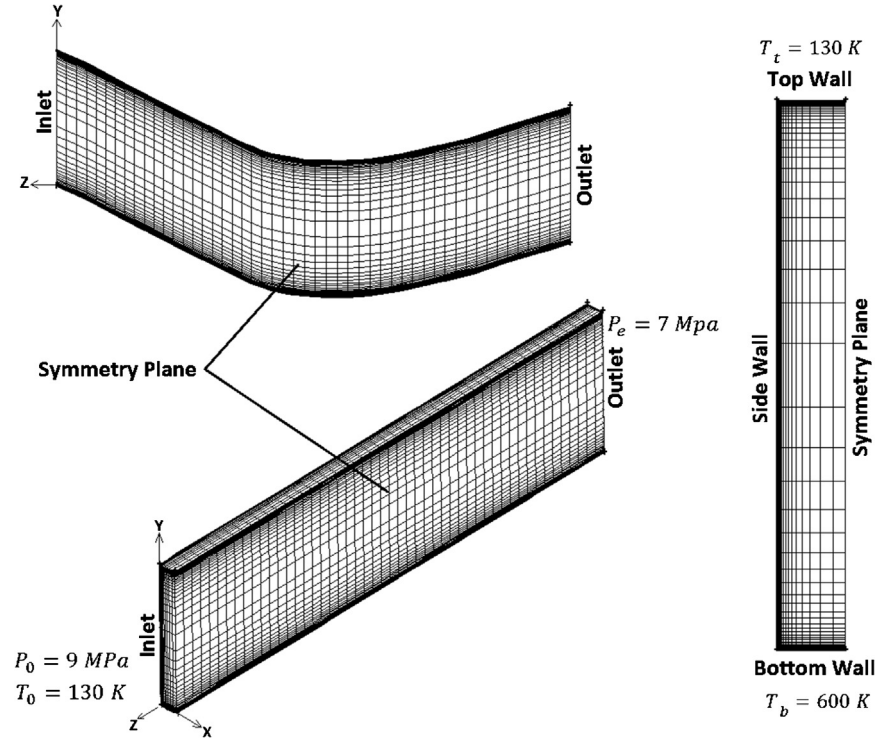


Fig. 5. Geometry, boundary conditions, and a schematic view of the computational mesh for the cooling channels (figures are not scaled).

Table 2

The sensitivity analysis of the numerical solution to the computational grid.

	Grid size	y^+	q_w (MW/m ²)
Coarse grid	$8 \times 23 \times 25$	4.193	5.16
Medium grid	$16 \times 46 \times 50$	1.281	7.93
Fine grid	$32 \times 92 \times 100$	0.963	8.97
Very fine grid	$64 \times 184 \times 200$	0.706	8.99

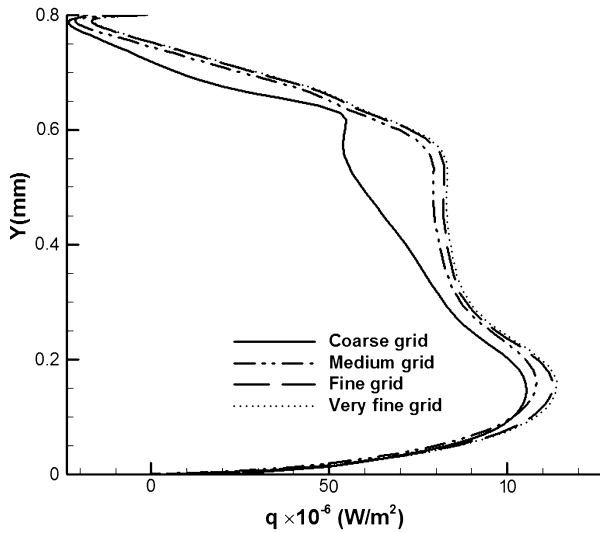


Fig. 6. Heat flux distribution in the side wall at the rectangular channel exit.

Schley [32], Quinones-Cisneros and Deiters et al. [31], and Younglove and Ely [28] are adopted. To better evaluate the accuracy of the utilized state and transport property relations in comparison with the Ref. [20], a solution with the S-A model is also considered. The wall temperature distribution in this solution, as well as the numerical simulation of Ref. [20], are compared to the experimental data in Fig. 7. The results of the present study show

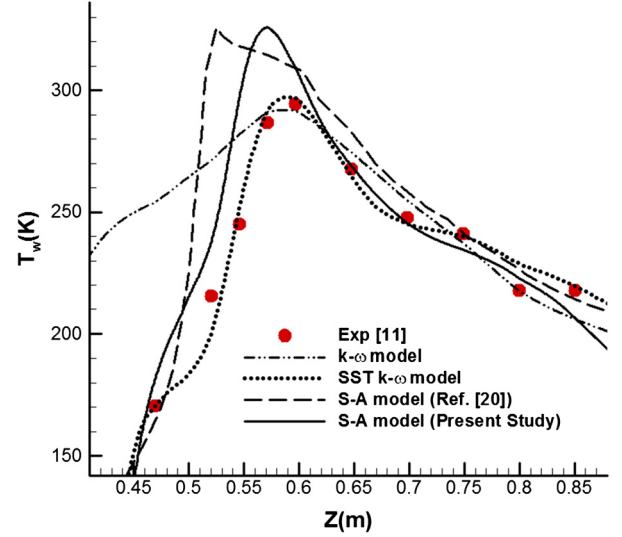


Fig. 7. Wall temperature distribution for different turbulence models and thermo-fluidic relations.

Table 3

Wall temperature errors of different turbulence models.

Turbulence model	AAD (%)
S-A model (present study)	6.9
S-A model (Ref. [20])	12
$K-\omega$	14.5
SST $K-\omega$	1.8

an error of around 5 percent less than Ref. [20], as reported in Table 3. Therefore, the thermo-fluidic relations employed in this study provide an acceptable accuracy for the transcritical coolant flow. These relations with the SST $K-\omega$ model are used in the rest of this study.

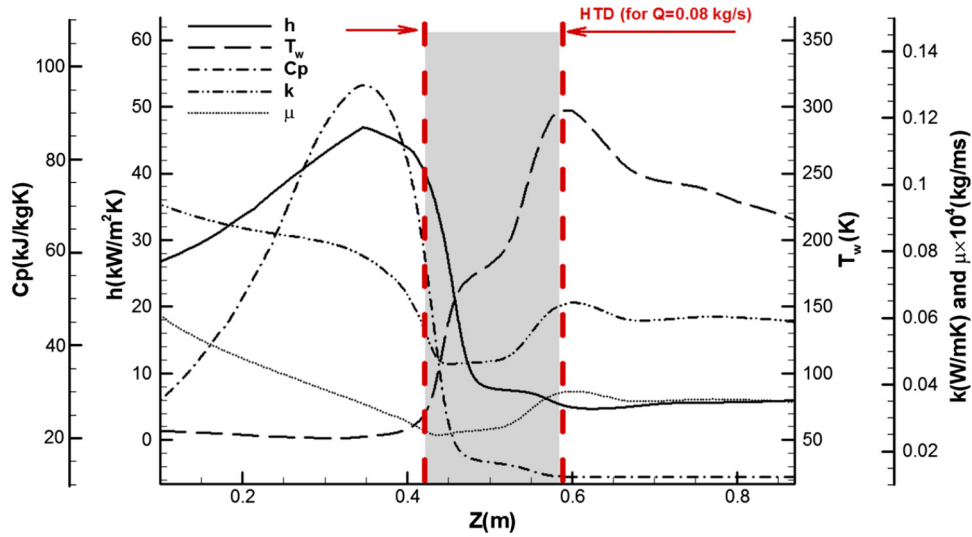


Fig. 8. The HTD and transcritical regions under the boundary conditions of Test Case 1.

6.1.2. Mass flow rate effect

In order to explore the region of coolant fluid state change (transcritical regime) and HTD phenomenon, the variations of the heat transfer coefficient, wall temperature, near-wall isobaric specific heat, thermal conductivity, and viscosity along the tube under the boundary conditions of Test Case 1 ($Q = 0.08$ kg/s) are illustrated in Fig. 8. The region in which C_p experiences a sharp change (the maximum value of C_p is about $Z = 340$ mm) indicates the location of the transcritical regime [16,19]. In this region the coolant state changes from liquid-like to gas-like. Generally, HTD is characterized by a severe reduction in the convective heat transfer coefficient and a peak in the wall temperature along the channel. Therefore, wall temperature (T_w), heat convection coefficient (h), and the wall heat flux (q) distributions are key parameters in HTD region recognition [15,20,21]. In the rest of this paper, T_w , h , k , and μ are used for determination of HTD region in Test Case 1 and q , h , k , and μ are adopted for Test Case 2. Fig. 8 shows that there are areas where the wall temperature encounters with the sharp local increment (i.e. the wall temperature increases from about 70 K to 290 K in the range of $Z = 420$ to 590 mm). Simultaneously, in this area, the thermal conductivity and viscosity reveal an increasing trend. This area depicts the HTD region in which the heat transfer coefficient decreases dramatically. Further, the region of fluid state change and HTD are relatively overlapped.

To study the effect of mass flow rate changes on the location of transcritical regime along the tube, the isobaric specific heat (C_p) distribution is studied in different mass flow rates. Fig. 9 shows the near-wall isobaric specific heat distribution for different mass flow rates from 0.08 to 0.11 kg/s. Also, the location of maximum C_p ($Z_{C_p, \max}$) is given in the figure legend. Increasing the mass flow rate results in an increment in $Z_{C_p, \max}$ that postpones the transcritical region and extends it along the tube. To consider the effect of mass flow rate changes on the HTD location, the variations of the wall temperature, near-wall thermal conductivity, and viscosity along the tube for different mass flow rates are illustrated in Figs. 10–12, respectively. By increasing mass flow rate, HTD is delayed with a lower intensity.

6.2. Methane in the straight rectangular channel

Transcritical methane flow in the high aspect ratio straight rectangular channel of Test Case 2 is studied. Moreover, the effect of bottom wall temperature, outlet pressure, and surface roughness, on pressure drop and the region of coolant fluid state change and HTD are discussed in this section.

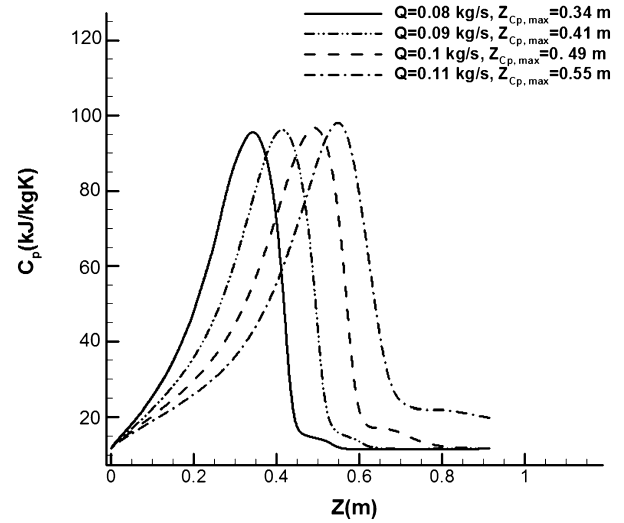


Fig. 9. Isobaric specific heat variations for different mass flow rates.

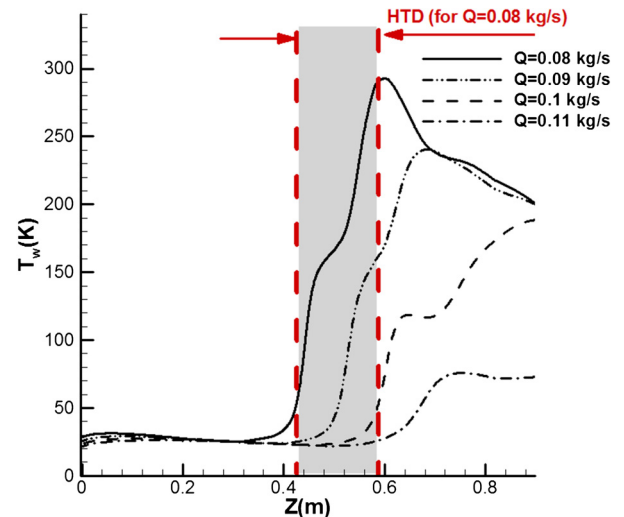


Fig. 10. Wall temperature distribution for different mass flow rates.

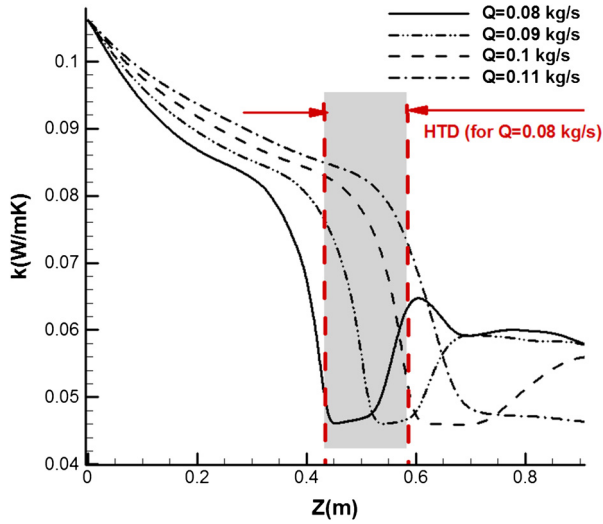


Fig. 11. Thermal conductivity variations along the wall for different mass flow rates.

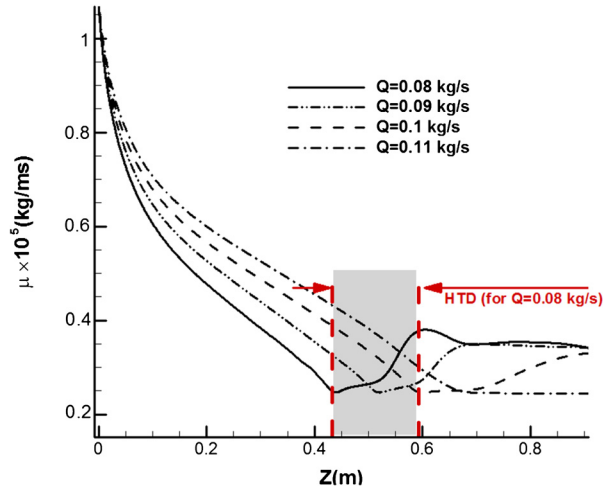


Fig. 12. Viscosity changes along the wall for different mass flow rates.

To investigate the region of coolant fluid state change (transcritical regime and HTD in the rectangular channel, the heat transfer coefficient, wall heat flux, isobaric specific heat, and thermal conductivity distributions at the channel outlet under the boundary conditions of Test Case 2 are presented in Fig. 13. The results show that the heat transfer coefficient and wall heat flux decrease and thermal conductivity increases above the $y \approx 0.58$ mm which indicates the HTD region. Further, the HTD occurs in the region of coolant fluid state change (i.e., the region where the isobaric specific heat has a maximum) which is consistent with studies by Refs. [14–16,19].

6.2.1. Bottom wall temperature effect

To consider the effect of bottom wall temperature on the thermal behavior of the transcritical coolant, three different temperatures are simulated. The methane density and velocity magnitude contours at the channel outlet for bottom wall temperatures of 200, 400 and 600 K are displayed in Fig. 14. Methane is in the liquid-like state for $T_b = 200$ K, whereas for 400 and 600 K temperatures both liquid-like and gas-like states exist concurrently. It can also be seen that increment of bottom wall temperature thickens the gas-layer. Moreover, the velocity magnitude of the coolant flow is higher near the bottom wall.

In order to investigate the effect of bottom wall temperature on the coolant fluid state change location, density and isobaric spe-

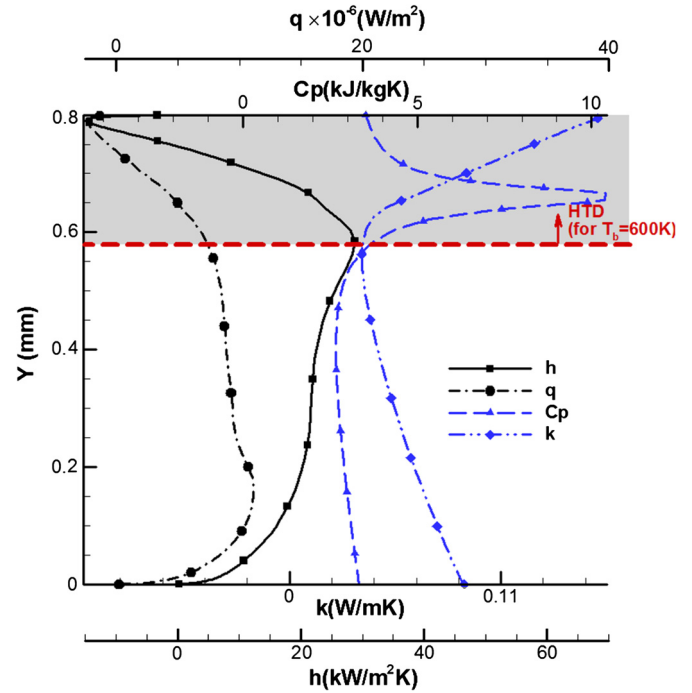


Fig. 13. The HTD and transcritical regions at the channel outlet under the boundary conditions of Test Case 2.

cific heat variations in the vicinity of the side wall at the outlet are sketched in Fig. 15. In this figure, density and isobaric specific heat transfer are colored in black and blue, respectively. High gradient of C_p diagram indicates the change region in the fluid state (shaded region on the figure for $T_b = 600$ K). This region is corresponding to the inflection point vicinity of density curve. Additionally, the fluid state change occurs at a higher distance from the bottom wall with increasing its temperature.

To investigate the effect of bottom wall temperature on the region of HTD phenomenon, the near-wall viscosity and wall heat flux distribution at the channel outlet are depicted in Fig. 16, wherein viscosity results are colored in blue and wall heat flux in black. In 400 and 600 K, viscosity decreases slightly up to a point and then suddenly raises with increasing height. The change of viscosity curve slope demonstrates the region of HTD in the rectangular channel. Furthermore, wall heat transfer is significantly decreased in this region. HTD occurs at higher distances from the bottom wall when its temperature increases (see Fig. 16). Moreover, the wall heat flux increases with increment in the bottom wall temperature. However, the heat flux reaches its maximum value adjacent to the bottom wall where a high-velocity gas layer exists (as seen in Fig. 14b) for $T_b = 400$ and 600 K.

6.2.2. Outlet pressure effect

Simulations are performed for three different outlet pressures to study its effect on the thermal behavior of the transcritical coolant. Fig. 17 indicates the effect of outlet pressure on the velocity magnitude contour at the channel outlet. For lower outlet pressures, velocity magnitude reaches higher values around the bottom wall. Also, the effect of outlet pressure on the region of HTD phenomenon is studied by plotting the near-wall thermal conductivity and the wall heat flux distributions at the channel outlet in Fig. 18. Reducing outlet pressure results in an increase in the wall heat flux (caused by the increment of the velocity in Fig. 17). Further, the location of conductivity diagram slope change, which is corresponding to the region of HTD, remains nearly fixed with outlet pressure changes. Additionally, the effect of outlet pressure on the location of fluid state change is studied in Fig. 19 in which density

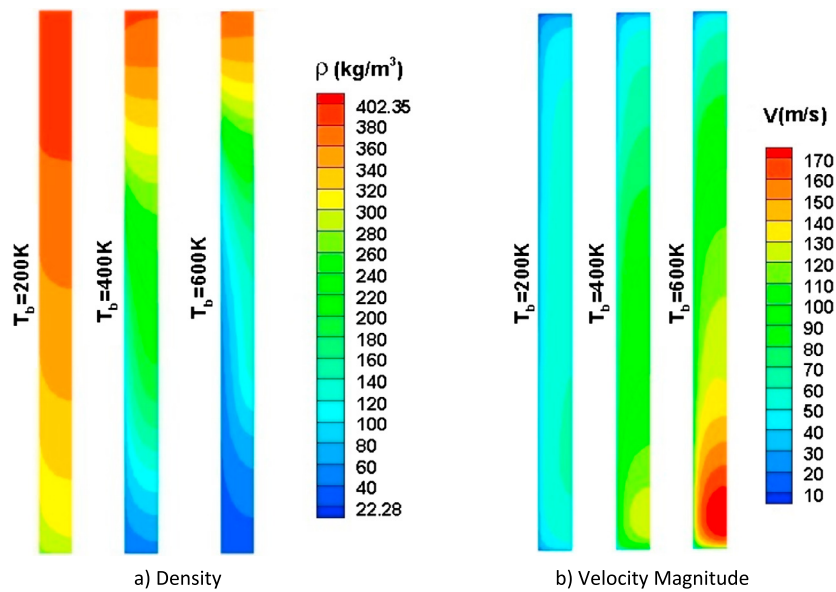


Fig. 14. Comparison of methane density (a) and velocity magnitude (b) contours at the channel outlet for different bottom wall temperature. (For interpretation of the references to color in this figure, the reader is referred to the web version of this article.)

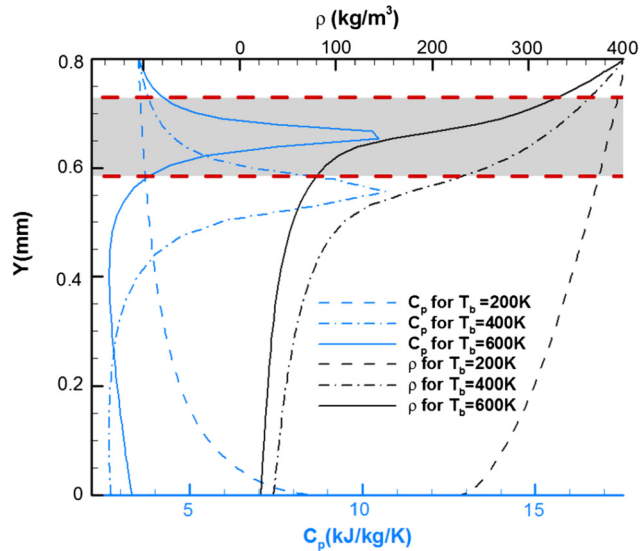


Fig. 15. Isobaric specific heat and density distributions near the side wall at the channel outlet for different bottom wall temperature.

Table 4
Stagnation pressure loss versus surface roughness.

Surface roughness [μm]	Outlet p_0 (bar)	Δp_0 (%)
Smooth	78.64	12.62
0.1	78.32	12.98
0.2	78.12	13.20
0.4	77.86	13.49
0.6	77.57	13.82

and specific heat diagram near the side wall at the channel outlet are illustrated. This figure indicates that the maximum value of isobaric specific heat is increased for lower outlet pressures; however, variations of the fluid state change region and the inflection point of density curve are inconsiderable.

6.2.3. Surface roughness effect

For the sake of investigating the effect of surface roughness on the thermal behavior of transcritical coolant, a copper alloy with

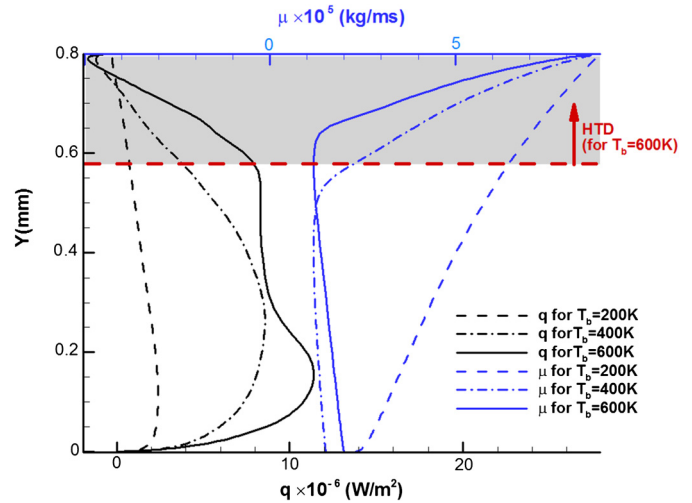


Fig. 16. Near-wall viscosity and wall heat flux distribution at the channel outlet for different bottom wall temperature.

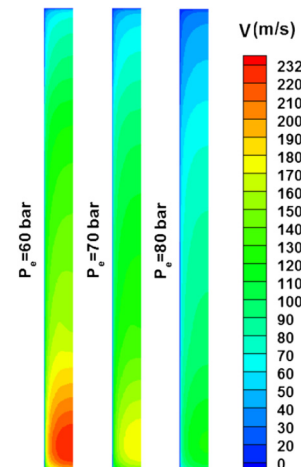


Fig. 17. Comparison of velocity magnitude contour at the channel outlet for different outlet pressures.

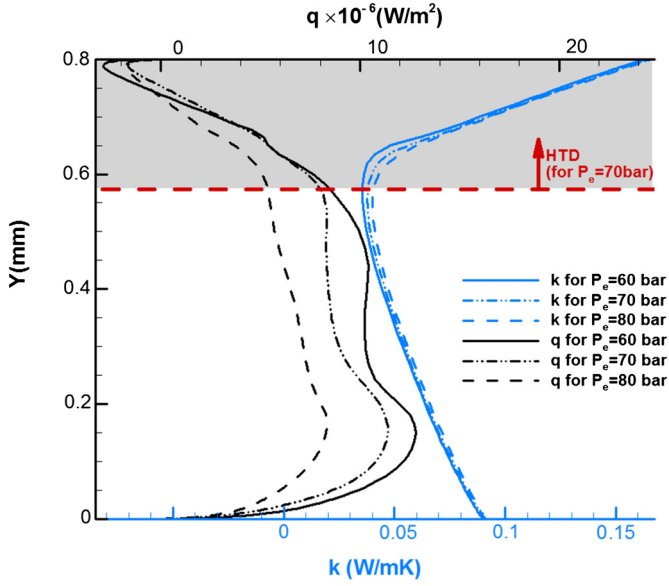


Fig. 18. Methane wall heat flux and near-wall thermal conductivity distribution at the channel outlet for different outlet pressures.

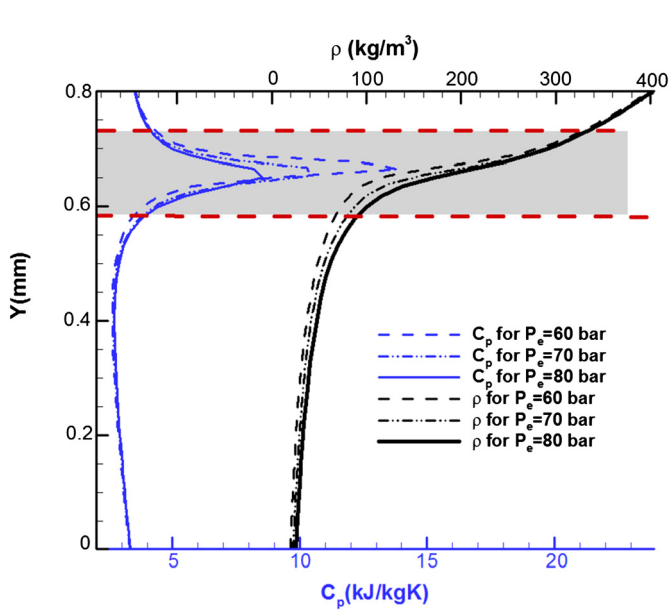


Fig. 19. Methane isobaric specific heat and density distribution near the side wall at channel outlet for different outlet pressures.

a surface roughness of 0.1 to 0.6 μm [12,23,41] is selected as the channel wall material. The total pressure at the inlet is set to be 90 bar. The results are given in Figs. 20 through 22. The stagnation pressure contours at the channel outlet are displayed in Fig. 20 for different levels of surface roughness. According to these results, stagnation pressure loss versus surface roughness is provided in Table 4. Increasing surface roughness causes 1.2% increment in stagnation pressure loss. Moreover, to study the effect of the surface roughness on the region of HTD phenomenon as well as the coolant fluid state change location, variations of density, wall heat flux, isobaric specific heat, and near wall viscosity are sketched in Figs. 21 and 22, respectively. As depicted in these figures, surface roughness affects the local maximum heat flux by improving thermal equilibrium; however, it has a negligible effect on both the region of HTD and fluid state change.

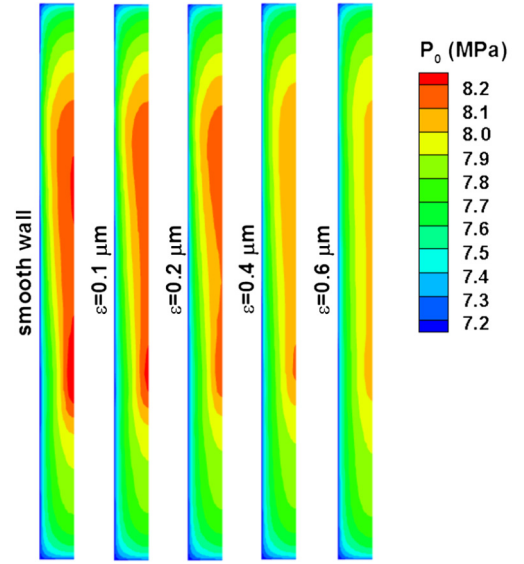


Fig. 20. Total pressure contour at the channel outlet for different surfaces roughness values.

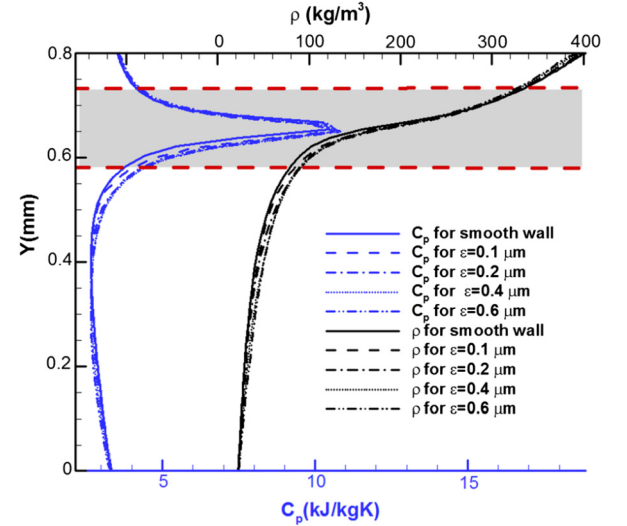


Fig. 21. Isobaric specific heat and density variations at the channel outlet for different surfaces roughness values.

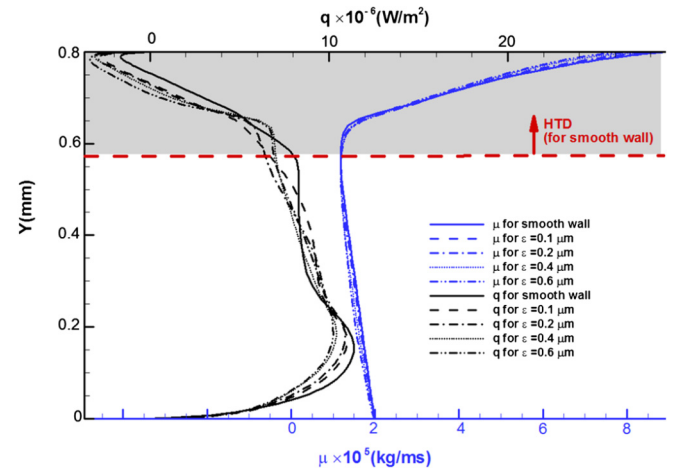


Fig. 22. Wall heat flux and near-wall viscosity variations near the side wall at channel outlet for different surfaces roughness values.

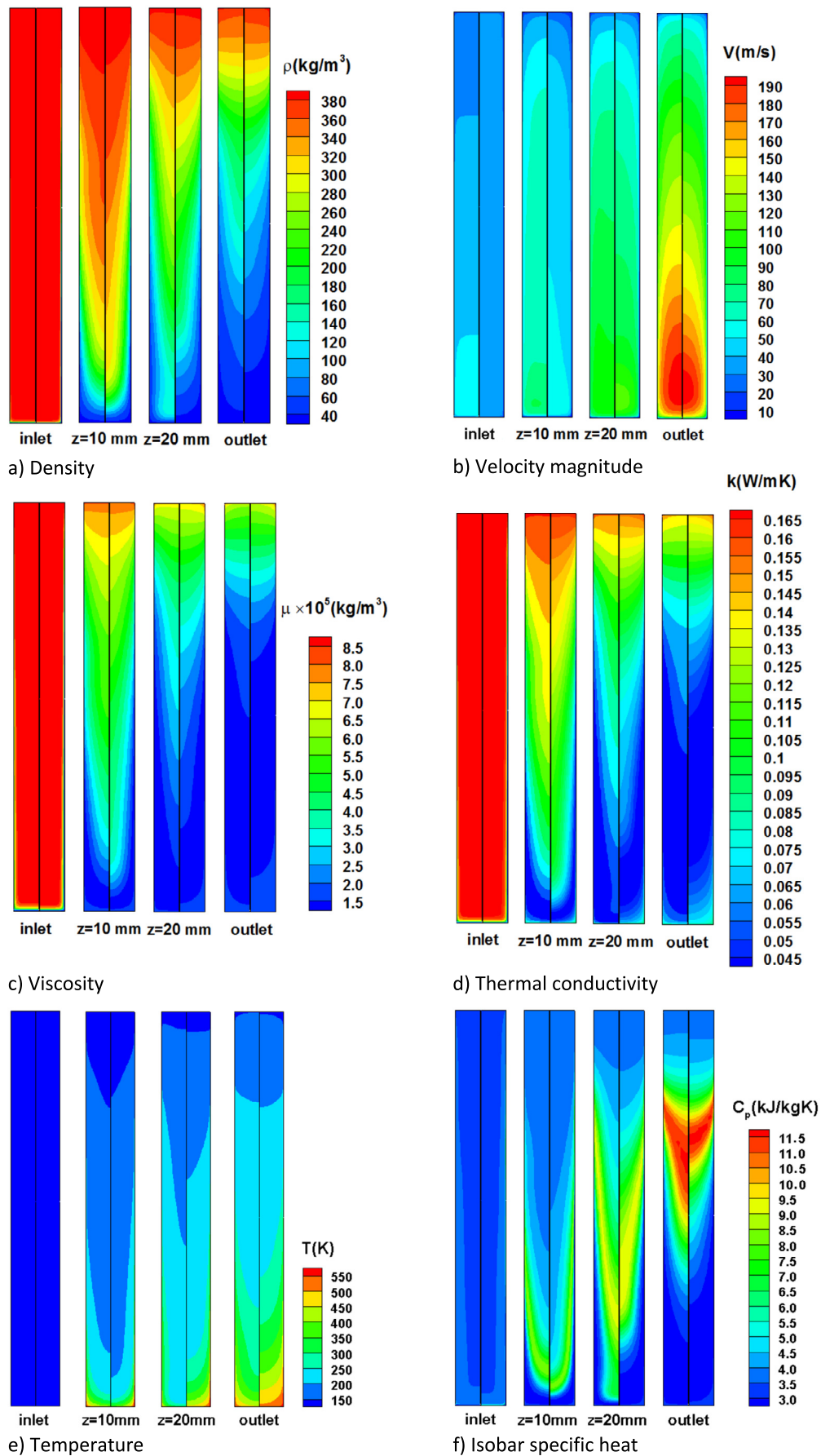


Fig. 23. Comparison of flow properties contours at different cross-sections of the straight and curved rectangular channels.

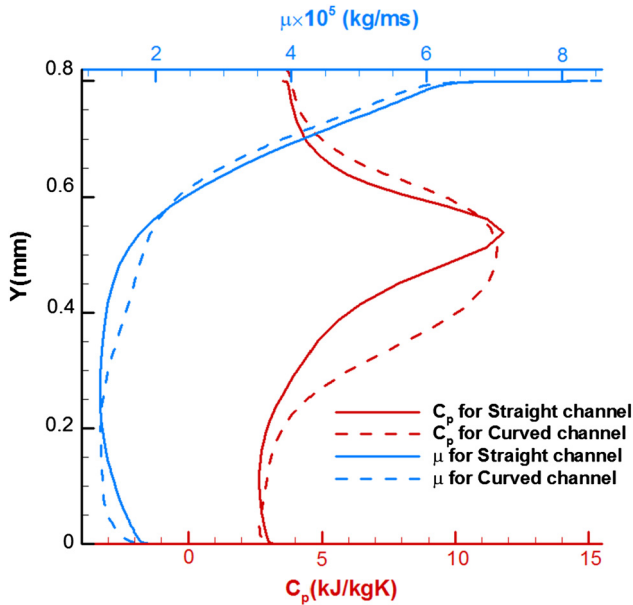


Fig. 24. Isobaric specific heat and viscosity distribution on the symmetry plane at the outlet of straight and curved channels.

6.3. Methane in the curved rectangular cooling channel

Methane is simulated in the curved rectangular channel (that is the case of Vulcain engine [23] in the vicinity of the throat), to study the effect of channel curvature on the thermal behavior of the transcritical coolant. Fig. 23 displays the contours of several flow parameters in both straight and curved channels at different cross-sections. In this figure, the right side of each cross section stands for the straight channel results, as the left side is for the curved channel. The results (Figs. 23a, and 23b) reveal that the average density decreases and the velocity magnitude increases along the channel as the gas layer emerges. The average density at the outlet of the curved channel is 1.3% higher than the straight channel, and velocity is 4% less. The curvature results in a thinner gas layer at the second half part of the channel. The results for μ , k , and temperature reveal to be less sensitive to the channel curvature as depicted in Figs. 23c, 23d, and 23e. Further, Fig. 23f shows that the average value of C_p increases along the channel as the coolant fluid absorbs heat. This value for the curved channel is about 12% more than the straight one at the outlet. The curvature leads to more extended C_p variations and fluid state change region at the second half part of the channel. Fig. 24 depicts the isobaric specific heat and viscosity on the symmetry plane at the outlet of the curved and straight channels to examine the effect of channel curvature on the region of HTD and fluid state change. As seen from the figure, the HTD occurs at a lower height in the curved channel at a certain cross-section and the fluid state change region is more extended as explained before.

7. Conclusions

A survey of the literature indicated that most previous numerical studies have focused on the HTD of the transcritical methane flow inside circular cooling channels. Whereas, the actual engine cooling channels are typically rectangular, curved, and heated asymmetrically. In addition, variations of the transport properties have not been investigated accurately along the cooling channel at near-critical conditions and deterioration of heat transfer. The purpose of the current article was to comprehensively study the thermal behavior, HTD, and fluid state change of the transcritical hydrogen and methane flow under the effect of different param-

eters inside a heated tube and straight/curved rectangular channels with asymmetric heating. The real gas state equations, the transport property relations, and turbulence models were examined for proper simulations. Three-dimensional analyses for the turbulent flow of the transcritical coolant and heat transfer were conducted. The accuracy of the present numerical approach was validated with numerical and experimental data. The main findings of the present paper are:

1. The region in which C_p experiences a sharp change, indicating the transcritical regime, is located in the vicinity of the inflection point of the density curve.
2. The change of thermal conductivity and viscosity curve slope is proposed as a criterion for locating the HTD region in which the wall temperature shows a local sharp increment.
3. Increasing the mass flow rate of the transcritical hydrogen inside a heated tube shifts the maximum value of C_p to farther downstream, which postpones and elongates the transcritical region along the tube. This also results in a delayed and less intensified HTD.
4. Increasing the bottom wall temperature of the straight rectangular channel thickens the gas-layer and also increases the velocity magnitude and wall heat flux of the transcritical methane. Moreover, the fluid state change and HTD region occur in a higher position.
5. The reduction of outlet pressure of methane flow leads to a velocity magnitude and heat flux increment near the bottom wall. Further, the location of a change in the conductivity diagram slope, which is corresponding to the HTD region, shows to be independent of the outlet pressure. The value of isobaric specific heat peak was increased with decreasing outlet pressure; while, the fluid state change region and the inflection point of density curve showed infinitesimal variations.
6. Surface roughness affects the local maximum heat flux due to the thermal equilibrium improvement. However, it has a negligible effect on both HTD region and the methane state change location.
7. The curvature of the channel has a direct effect on the gas layer thickness and C_p at the second half of the channel. Further, the HTD occurs at a lower height and the fluid state change region is more extended.

Conflict of interest statement

The authors have no conflict of interest to declare.

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